THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** push or pull	Result of interaction between objects
2	**(d)** shape	Push changes shape
3	**(d)** muscular force	A contact force
4	**(d)** friction due to air	Fluids exert friction
5	**(c)** electrostatic force	Static charges
6	**(c)** gravitational force	Acts downwards
7	**(a)** mass	Matter does not change
8	**(a)** spring balance	Stretches a spring
9	**(d)** upthrust (buoyant force)	Opposes weight
10	**(a)** has trapped air and is less dense than water	Porous volcanic rock

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A push or pull from interaction between objects (1). SI unit is newton (1).	1+1
12	Move a resting object, change speed (1). Change direction, change shape (1).	1+1
13	Contact forces act on touching objects, e.g. friction (1). Non-contact forces act without contact, e.g. gravity (1).	1+1
14	Force due to action of muscles (1). Walking and lifting (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A friend holding your bicycle from behind pulls it (1½). The speed decreases, showing force changes speed (1½).	1½+1½
16	Start: kicking a football (1). Change shape: pressing a balloon (1). Change direction: bat hitting ball (1).	1+1+1
17	Several forces may act (1). They balance one another (1). Net force is zero (1).	1+1+1
	OR: Surfaces have minute irregularities (1). They lock into each other in contact (1). This opposes motion (1).	1+1+1
18	Streamlined shape (1). Smooth surface (1). Reduces resistance of air or water (1).	1+1+1
	OR: Rub scale with dry hair or cloth (1). Bring it near paper bits (1). Paper is attracted by static charge (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1). Move from rest (1). Change speed (1). Change direction (1). Change shape (1).	1+1+1+1+1
	OR: Contact forces act on touching objects (1). Muscular force in walking and chewing (2). Friction opposes motion due to surface irregularities (2).	1+2+2
20	Non-contact forces act without touching (1). Magnetic – magnets attract or repel (1½). Electrostatic – charged scale attracts paper (1½). Gravitational – falling objects (1).	1+1½+1½+1
	OR: Mass is matter in kg (1). Weight is Earth's pull in N (1). Earth: 50 N (1). Jupiter: 5 × 25.4 = 127 N (1). Mass same on both (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** newton	Symbol N
2	**(c)** direction of motion	Force changes motion
3	**(c)** friction	Acts opposite to motion
4	**(b)** gravitational force	Acts without touching
5	**(b)** static charges	Build up on rubbing
6	**(b)** force with which Earth pulls it	Measured in newtons
7	**(c)** 10 N	As per the textbook values
8	**(c)** maximum weight it can measure	e.g. 0 to 10 N
9	**(b)** equal to buoyant force	Balanced forces
10	**(b)** upthrust	Liquid pushes it up

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force that comes into play when an object moves or tries to move over another surface (1). Opposite to the direction of motion (1).	1+1
12	Rough surface has more irregularities (1). So friction is greater (1).	1+1
13	Can attract or repel (1). Acts without contact (1).	1+1
14	They repel each other (1). This is electrostatic force (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnetic: attract and repel, magnets (1). Electrostatic: attract and repel, charges (1). Gravity: attractive only, all masses (1).	1+1+1
16	It rises and slows down (1). Stops momentarily at top (1). Then falls with increasing speed due to gravity (1).	1+1+1
17	Earth: 60 × 10 = 600 N (1½). Moon: 60 × 1.6 = 96 N (1½).	1½+1½
	OR: Mass = 50 ÷ 10 (1½). 5 kg (1½).	1½+1½
18	Least count = 1 ÷ 10 = 0.1 N (1½). Reading = 4 + 0.3 = 4.3 N (1½).	1½+1½
	OR: kg is a unit of mass (1). Weight is a force measured in newtons (1). Correct statement: mass is 10 kg (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Spring stretches under load (1). Scale in newton (1). Least count = difference between marks ÷ number of divisions (1). Example 1 N ÷ 5 = 0.2 N (1). Reading with a numerical example (1).	1+1+1+1+1
	OR: Gravity pulls down, upthrust pushes up (1). Float when balanced (1). Sink when gravity greater (1). Principle statement (1). Applications: ships, pumice, boats (1).	1+1+1+1+1
20	Rub two balloons on hair (1). They repel each other (1). Rubbed balloon attracts uncharged paper (1). Like charges repel, unlike attract (1). Charged object attracts uncharged object (1).	1+1+1+1+1
	OR: Useful: walking, holding, brakes (2). Harmful: wear and tear, slows moving parts (1). Reduce by smooth surfaces and lubricants (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.